Teaching a Neural Network the Glow of the Solar System

A geometrically constrained model of zodiacal light for NASA's SPHEREx mission

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A geometrically constrained neural network learns the zodiacal light from viewing geometry alone, reaching ~0.03% median error on months it never saw.

Background

SPHEREx is a NASA space telescope, launched in March 2025, that maps the **entire sky across a range of infrared wavelengths** (0.75–5 μm). Its maps will help explain how the universe began, how galaxies formed, and where water and other ingredients for life are found.

Zodiacal light. The Solar System is filled with a thin cloud of dust from comets and asteroids. Sunlight scattered by this dust, plus the dust's own heat glow, adds a smooth glow across every SPHEREx image (Fig. 1). It is bright enough to be confused with the faint background light from distant galaxies, and it changes with where the telescope looks and where Earth is in its orbit, so it must be carefully modeled and removed.

The Kelsall model. Kelsall et al. (1998) built the standard model of this dust from NASA's COBE satellite: a smooth cloud, three dust bands, a ring around the Sun and a clump trailing Earth. The software **ZodiPy** computes what a telescope would see from it; we use it to simulate SPHEREx data.

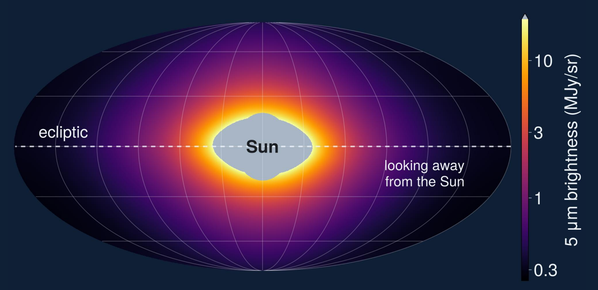


Figure 1. The whole sky at 5 μm as seen from SPHEREx (ZodiPy). The glow is brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

The problem: zodiacal light must be removed from every SPHEREx image, but physics models such as Kelsall's were fit to older data and leave errors behind.

Research objective: build a **geometrically constrained** neural network that learns the zodiacal light directly from SPHEREx data and **outperforms the Kelsall model**. First step (this poster): show that it can learn ZodiPy's simulated sky from geometry alone.

Methods

A geometrically constrained model. The network never sees a date or a position on a sky map. Its only inputs are 9 numbers describing the viewing geometry (Fig. 2): dot products of SPHEREx's look direction n with the directions that shape the zodiacal light, plus Earth's place in its orbit. It must learn how the brightness depends on geometry, just as the physics does, instead of memorizing when and where SPHEREx looked.

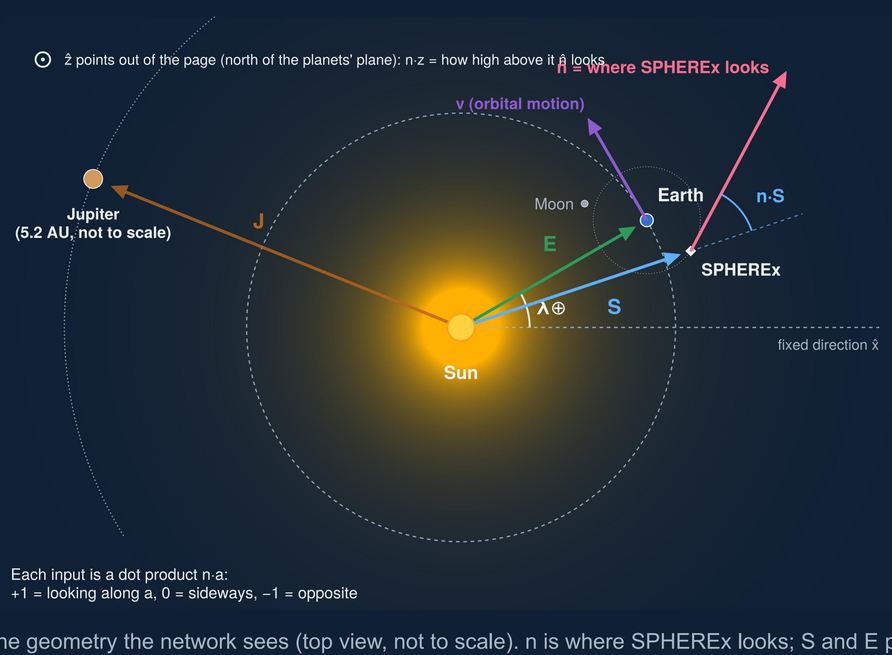


Figure 2. The geometry the network sees (top view, not to scale). n is where SPHEREx looks; S and E point from the Sun to SPHEREx and Earth; v is the orbital motion; λ⊕ is Earth's angle around its orbit. Jupiter and the Moon are shown for context: the final model uses only the Sun–Earth–SPHEREx geometry and the time of year.

The 9 inputs

n·S, n·E	toward or away from the Sun (seen from SPHEREx and from Earth)
n·z	how far above or below the plane of the planets
n·vS, n·vE	ahead of or behind the orbital motion (SPHEREx, Earth)
n·x⊕, n·y⊕	look direction in a frame that turns with Earth (x⊕ = Sun → Earth)
sin λ⊕, cos λ⊕	Earth's position in its orbit: the time of year

Training and testing. ZodiPy simulates 423,607 SPHEREx pointings (140 days, 5 μm). Five small networks (4 hidden layers, 64 → 16 neurons; Flux.jl in Julia) learn brightness from the 9 inputs and are averaged, then tested on whole months and sky regions they never saw.

Lesson: adding the time of year (sin λ⊕, cos λ⊕) cut the errors about 40–65×.

Results

0.026% median error on months withheld from training

95% of pointings within 0.1% of ZodiPy

≤0.04% median error for every choice of withheld months

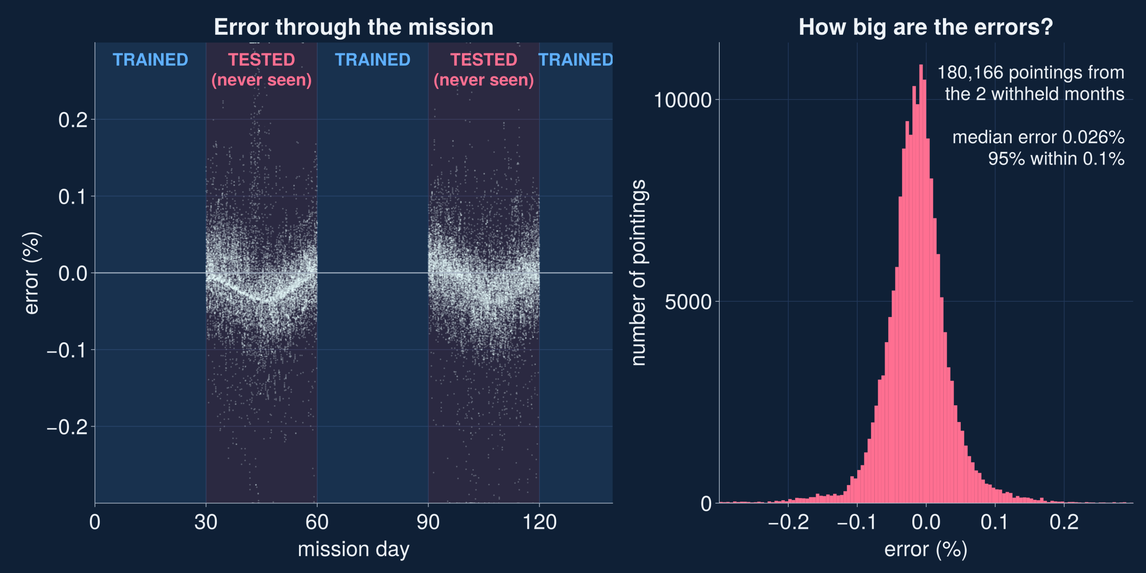


Figure 3. Neural network vs. ZodiPy. The network was trained on mission days 0–30, 60–90 and 120–140, then tested on the two 30-day blocks it never saw (days 30–60 and 90–120; 180,166 pointings). Half of the errors are below 0.026% and 95% are below 0.1%. (With a random 10% of pointings withheld instead, the median is 0.015%.)

Where on the sky are the errors? We repeated the test six times, sliding which months were withheld, so almost every pointing was tested by networks that never saw it. Every version reached 0.023–0.037% median error. The errors are largest along the plane of the planets, where the dust bands are (0.09%), and smallest toward the poles (0.016%).

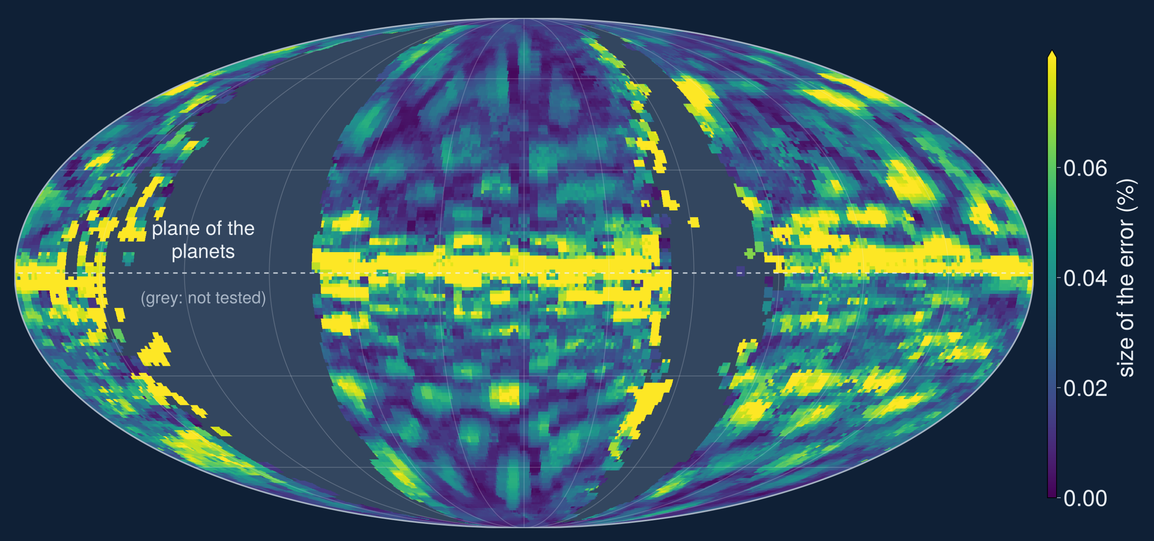


Figure 4. The network's error on pointings it never saw, across the whole sky (plane of the planets across the middle). Each pointing is drawn as a 3.5°-wide patch, about the width of SPHEREx's view. Grey: not observed in the 140 days, or only at the very start or end of the mission.

Next Steps

✓ DONE Validate	Learn ZodiPy's simulated sky from geometry alone; test on unseen months and sky regions
IN PROGRESS Refine	Longer training; all SPHEREx wavelengths (a first 1.25–5 μm version: ~0.1% error)
NEXT Scale up	Move training to a computing cluster
NEXT Real data	Train on real SPHEREx images and compare with the Kelsall model

All results so far are preliminary and use simulated data.

Acknowledgements

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References

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